José Manuel Requena Plens

Firmware & Software Engineer

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On-site, remote or hybrid

Summary

Embedded firmware and software engineer with experience building industrial firmware for solar inverters (C, STM32, FreeRTOS) and a solid R&D background in acoustics (physics). Focused on low-level optimization, industrial communications (Modbus) and hardware test automation. I maintain *open source* projects used by developers worldwide. Targeting **firmware, software and QA** roles, bringing engineering rigor and a focus on quality.

Experience

R&D Software Developer **Apr 2023 - May 2026**

Power Electronics S.L. (PE) — Solar + Storage (PE Solar) · Lliria, Spain

Firmware development for solar inverters and control systems in photovoltaic plants, working on STM32 microcontrollers with bare-metal and RTOS environments.

- Reduced RTOS task stack usage by 30% to 50%, optimizing memory on STM32 microcontrollers.
- Optimized Modbus TCP request handling, cutting CPU usage under load (40,000 req/s) from 80% to 17% with no dropped responses.
- Redesigned embedded file-system access, cutting access times by over 50%.
- Implemented a scheduled purge (daily/monthly) that safely clears thousands of files, preventing storage exhaustion under continuous operation.
- Built and maintained industrial communications in C with hardware abstraction layers (HAL): Modbus RTU/TCP, UART, I2C and SPI.
- Automated the test suite: ~50 to 80 unit tests per module (Ceedling, thousands in total) reaching 100% line and branch coverage via standard-library mocks/fakes, plus hundreds of on-hardware integration tests (pytest) for TCP/FTP performance and FTP security, all in CI.
- Reviewed merge requests across a 12-person team, catching bugs and proposing improvements before integration.

Predoctoral researcher **Jan 2023 - Apr 2023**

Polytechnic University of Valencia (UPV) · Spanish National Research Council (CSIC) · Hospital La Fe (IIS) — Medical Imaging and Therapy Systems (i3M) · Valencia, Spain

Researcher in the R&D line: **Acoustic Metamaterials for Ultrasound Histotripsy treatments**. **Project**: New generation of smart metasurfaces based on additive manufacturing for strategic applications in telecommunications (Metasmart). Funded by the Valencian Agency for Innovation (Reference: INNEST/2022/345).

- Ultra-close focus acoustic lens design.
- Prototype manufacturing.
- Experimentation with ex-vivo tissues.

Predoctoral researcher **Jul 2021 - Jul 2022**

Polytechnic University of Valencia (UPV) · Spanish National Research Council (CSIC) — Medical Imaging and Therapy Systems (i3M) · Valencia, Spain

The main objective is to conduct research and development of acoustic metamaterials for use in architectural applications and medical imaging.

- Developed acoustic metadiffusers reaching a diffusion coefficient of 0.8 with a thickness 10× smaller than traditional diffusers.
- Simulation of the acoustic diffusers and validation of results (MATLAB & COMSOL).
- Designed and 3D-printed (resin) slitted-tube waveguides with numerically-computed slits (2 to 3 cm tubes, ~0.5 mm slits) that increased ultrasound directivity and range ~15× in air, validated experimentally.
- Validated experimental results against those obtained in simulation.
- Advisor of the Master thesis of a student of the Master's Degree in Acoustic Engineering of the UPV.

Researcher in European project **Feb 2020 - Jul 2021**

Higher Polytechnic School of Gandia (UPV) · European Space Agency (ESA) — Complex Media Acoustics Group of IGIC (IGIC) · Gandia, Spain

Implement and experimentally validate a sound mitigation method, applicable to a real space vehicle launch configuration (VEGA), that results in a significant decrease in sound pressure levels generated in the launch area during spacecraft liftoff. Project funded by ESA (European Space Agency) with reference ESA AO/1-9479/18/NL/LvH, in collaboration with: CNRS/Laboratoire d'Acoustique de la Université du Mans; COMET Ingeniería; Spanish National Research Council (CSIC) and Polytechnic University of Madrid.

- Designed the optimum acoustic-metamaterial geometry, achieving near-total absorption ($\alpha \approx 1$) across 200 to 500 Hz with a subwavelength design (wavelength/thickness ratio ≈ 16.5 vs ~2 conventional), validated on real lab prototypes.
- Simulation of a real environment to validate design performance (MATLAB, COMSOL & Python).
- Designed and 3D-printed the first prototype; for the large-scale version designed the injection-molding model and sourced the manufacturing of 600 modules with a supplier.
- Development of software for acoustic measurements according to ISO 10534-2 and ASTM E2611 standards. A|Lab
- Design and development of a 3-axis robotic system to perform experimental measurements. Photos and some information here: NEWS
- Results validated and accepted by ESA.

Research internship **Feb 2018 - Jul 2018**

University of Alicante (UA) — Dep. of Physics, Systems Engineering and Signal Theory (DFESTS) · Alicante, Spain

Internship in different research projects in the Applied Acoustics Group, belonging to the University Institute of Physics Applied to Sciences and Technologies of the Higher Polytechnic School of Alicante.

- Simulation of acoustic models.
- Acoustic measurements by means of a LabView-controlled robotic system.
- Investigation of the acoustic radiation efficiency (Vibration-borne noise) of metallic plates for an initial study of ship noise emissions in water. Carried out in collaboration with SAES, here you can read a news item on the subject: NEWS.

Founder and Vocal Member **Sep 2016 - Jul 2018**

AmicroTech (AμTech) — University of Alicante (UA) · Alicante, Spain

Association based at the Higher Polytechnic School of the University of Alicante. Promotes project-oriented microcontroller programming knowledge.

- Taught 30+ students in reinforcement and extension classes for the subject "Digital Electronic Systems" of the Degree in Sound and Image Engineering of the University of Alicante.
- Laboratory organization and planning (located at the Colegio Mayor).
- Communication and Public Relations.

Sound technician **Jun 2011 - Feb 2018**
Acusticox · Cox, Spain

- Equipment installation and adjustment for events.
- Equipment installation and adjustment for permanent installation.
- FOH Technician.
- Customer acquisition.

Laboratory technician **Sep 2005 - Nov 2009**
Profdent · Alicante, Spain

- Reception of work and casting of plaster models from impressions (alginate/silicone), with and without implants.
- Mounting articulators and casting metal crowns in a centrifuge.
- Ceramic application on crowns and bridges.

Technical Skills

Embedded Firmware & Electronics: C, STM32 & ARM, ESP32 / ESP-IDF, PlatformIO, Texas Instruments (MSP430, Tiva C, Code Composer Studio), FreeRTOS / RTOS, Comm. Protocols (I2C, SPI, RS485, Modbus), HW Debugging (JTAG / SWD), Keil uVision, Instrumentation (Oscilloscopes, DAQs, Logic A.), Secure Boot & Embedded Security (secure boot, flash encryption, key derivation), Doxygen (firmware documentation)

Software Engineering & Tools: Python, C++, Go, TypeScript, Bash (shellcheck, bats), Clang Tooling (clangd, clang-format, clang-tidy), Git, GitHub, GitLab, Docker, MATLAB, CMake, JetBrains Toolbox (CLion, PyCharm, WebStorm)

Quality, CI/CD & DevSecOps: GitHub Actions, GitLab CI, Jenkins, SonarQube / SonarCloud, GoReleaser (multi-platform releases), Static Analysis & SAST (CodeQL, gitleaks, golangci-lint), Testing (Ceedling, pytest, Playwright, Vitest, bats), pre-commit hooks

Networking: TCP/IP & Networking, MikroTik RouterOS, WireGuard / VPN, IPv6 & Dual-stack, QUIC / HTTP3, DNS / Dynamic DNS

Security & Cryptography: Applied Cryptography (AEAD, encrypt-then-MAC, KDFs), TLS / mTLS / PKI (client certificates), Nginx Hardening (CSP, security headers), CrowdSec (IDS/IPS), Network Hardening (firewall, tarpit, honeypot)

AI & Agentic Engineering: Model Context Protocol (MCP) (Server development in Go), AI Coding Agents (Claude Code, Copilot, agent skills/workflows), LLM Integration (API, prompt engineering, tool-use)

Simulation & Acoustics: COMSOL Multiphysics, LabView, Acoustic Instrumentation (laser/vibration accelerometers, microphones)

General & Platforms: Linux / Unix, MacOS, Windows, LaTeX, HTML / CSS

Open Source Projects

gitlab-mcp-server — Go · Model Context Protocol server
49 stars • 59 releases • ~173k downloads • On npm, PyPI & NuGet; listed on Glama, mcp.so, Cursor & mcpservers.org
Open-source GitLab MCP server that exposes 850+ GitLab actions (1,000+ depending on edition and deployment) to AI assistants through a dynamic 2-tool find/execute design, with stdio/HTTP/OAuth transports and safe & read-only modes.

Kleidos — C++ · ESP-IDF · Embedded security
In development
Hardware password manager firmware for nine M5Stack and LilyGO ESP32 boards: credentials stay encrypted on the device (AES-256-CBC with HMAC-SHA256, PBKDF2) and are typed over Bluetooth LE as a HID keyboard after an on-device unlock, with no app and no cloud.

phonometry — Python · DSP · Acoustics
133 stars • 28 releases • PyPI
Standards-conformant acoustics library: sound level metrology, psychoacoustics, room, building, environmental and underwater acoustics, vibration, electroacoustics and wave simulation, every method implemented from the standard, with 1897 CI-enforced conformance checks across 111 domains, against 227 normative designations and 125 further published sources. Formerly PyOctaveBand.

cs-routeros-bouncer — Go · Firewall · MikroTik · WAF
24 stars • 22 releases • On the CrowdSec Hub
CrowdSec bouncer for MikroTik RouterOS: feeds the collaborative IPS/WAF CrowdSec decisions into the RouterOS firewall (address lists) through its API, blocking malicious IPs in real time.

TFG-TFM_EPS — LaTeX · Academic template
84 stars • 6 releases
LaTeX template for Bachelor's and Master's theses at the University of Alicante (EPS), with an automated CI build pipeline.

Education

PhD program in Health and Wellness Technologies (discontinued) **2019 - 2023**
Polytechnic University of Valencia (UPV) — Ultrasound Medical and Industrial Laboratory (UMIL) · Valencia, Spain
Before completing the doctorate, I redirected my career toward industrial software development, applying my foundation in data analysis, programming and analytical problem-solving. The discontinued program focused on industrial and biomedical applications of metamaterials for the control of acoustic beams, combining new air-coupled ultrasound transducers with hyperbolic metamaterials to design vision systems that exceed the imaging capabilities of traditional systems.

- Biomedical signal processing and acoustic analysis algorithms.
- Ultrasound characterization for medical applications.
- Computational modeling and simulation of acoustic phenomena.
- Development of experimental methodologies for biomedical studies.
- Publication of results in specialized conferences and journals.

Master's Degree in Acoustical Engineering **2019**
Polytechnic University of Valencia (UPV) — Higher Polytechnic School of Gandia (EPSG) · Gandia, Spain
Degree obtained with honors in: Fundamentals of Acoustics, Acoustic Isolation, Musical Acoustics, Signal Processing in Acoustic Engineering, Ultrasonics and Acoustic Simulation Techniques. Final Master's thesis entitled **Difusores acústicos basados en resonadores de membrana y placa**

('Acoustic diffusers based on membrane and plate resonators'), qualified with honors with special mention.

Degree in Telecommunication Engineering in Sound and Image **2018**
University of Alicante (UA) — Higher Polytechnic School (EPS) · Alicante, Spain
Degree obtained together with the 2 specializations: Acoustic Engineering and Audiovisual Technology. Final Degree Work entitled Estudio de la relación campo directo/reverberado; útil/perjudicial ('Study of the direct/reverberated field relationship; useful/harmful').

Superior Sound Technician (Higher Vocational Degree) **2011**
IES Luis García Berlanga · Alicante, Spain

Electronics Technician (Intermediate Vocational Degree) **2009**
IES Salesianos (San José Artesano) · Elche, Spain

Certificates

Occupational Health and Safety

- Generic (INVASSAT, 50 Hours)
- Nanomaterials (INVASSAT, 50 Hours)
- Chemical sector (INVASSAT, 50 Hours)
- Emergencies (INVASSAT, 70 Hours)
- Nutrition (INVASSAT, 50 Hours)
- Educational (INVASSAT, 50 Hours)
- Services (INVASSAT, 50 Hours)
- Researcher (UPV, 15 Hours)

Transversal competences

- Gender perspective (EVES, 20 Hours)
- Teamwork (Labora, 25 Hours)
- Design Thinking (Labora, 25 Hours)
- Critical Thinking (Labora, 25 Hours)
- Adaptability, flexibility and agility (Labora, 25 Hours)
- Workplace autonomy and innovation, initiative and creativity (Labora, 25 Hours)
- Improving professional efficiency (Labora, 25 Hours)
- Entrepreneur gender perspective (UPV, 20 Hours)

Labor/Industrial

- Op. Industrial trucks (Gescoform, 15 Hours)
- Food Hygiene Course (Asonoman, 30 Hours)
- Self-protection plans (INVASSAT, 15 Hours)
- Static electricity (INVASSAT, 15 Hours)
- Data privacy (CSIRT-CV, 10 Hours)

Information technology

- Using Python for Research (HarvardX, 50 Hours)
- Analyzing Data With Python (IBM, 20 Hours)
- Visualizing Data with Python (IBM, 20 Hours)

Publications & conferences

Journal

- Francisco Castells, **José M. Requena-Plens** (2019). Loudspeakers for Vented Enclosures: a Backwards Approach for Speaker Selection (LoVE BASS). *Voice Coil*.

Conferences

- **José M. Requena-Plens**, Rubén Picó, Víctor J. Sánchez-Morcillo, Noé Jiménez, Alejandro Cebrecos, Mara Salut Escartí-Guillem (2021). Perfect broadband sound absorber metamaterial for noise reduction in a rocket launch. *European Congress and Exposition on Noise Control Engineering (EuroNoise 2021)*.
- **José M. Requena-Plens**, Jean-Philippe Groby, Noé Jiménez, Vicente Romero-García (2021). Sound diffusing metasurfaces based on elastic plates and membranes. *European Congress and Exposition on Noise Control Engineering (EuroNoise 2021)*.
- Mara Salut Escartí-Guillem, Pablo Barriuso Feijoo, Alejandro Cebrecos, Marcos Chimeno Manguán, Pedro Cobo Parra, Luis Miguel García Raffi, Jean-Philippe Groby, Sergio Hoyas Calvo, Noé Jiménez, Mario Lázaro Navarro, Julien Leng, José Nieto Mocholí, Rubén Picó, **José M. Requena-Plens**, Elena Roibás Millán (2021). Application of metamaterials to control noise scattering during space vehicle lift-off. *European Congress and Exposition on Noise Control Engineering (EuroNoise 2021)*.
- Mara Salut Escartí-Guillem, **José M. Requena-Plens**, Pablo Barriuso Feijoo, Alejandro Cebrecos, Marcos Chimeno Manguán, Pedro Cobo Parra, Luis Miguel García Raffi, Jean-Philippe Groby, Sergio Hoyas Calvo, Noé Jiménez, Mario Lázaro Navarro, Julien Leng, José Nieto Mocholí, Elena Roibás Millán, Vicente Romero-García, Rubén Picó, Víctor J. Sánchez-Morcillo, F. Simón Hidalgo, I. C.S. Ngan (2021). Launch sound level characterisation and mitigation: numerical modelling framework and metamaterial proof of concept. *16th European Conference on Spacecraft structures, Materials & Environmental Testing*.
- **José M. Requena-Plens**, Noé Jiménez, Alejandro Cebrecos, Rubén Picó, Víctor J. Sánchez-Morcillo (2020). Acoustic field prediction during the launch of rockets. *Tecniacústica 2020: 50º Congreso Español de Acústica. XI Congreso Ibérico de Acústica. Faro, Portugal*.
- Noé Jiménez, Trevor J. Cox, **José M. Requena-Plens**, Eric Ballestero, Jean-Philippe Groby, Vicente Romero-García (2020). Beyond Schroeder diffusers using acoustic metasurfaces. *Tecniacústica 2020: 50º Congreso Español de Acústica. XI Congreso Ibérico de Acústica. Faro, Portugal*.
- Jaime Ramis, Jesús Carbajo, Juan de Dios González, Pedro Poveda, **José M. Requena-Plens**, Enrique G. Segovia, others (2018). Aprendizaje basado en proyectos en las materias transductores acústicos y vibroacústica. *Memorias del Programa de Redes-ICE de calidad, innovación e investigación en docencia universitaria. Convocatoria 2017-18*.
- **José M. Requena-Plens**, Jenaro Vera Guarinos (2018). Cálculo corregido, basado en la teoría moderna, de los campos acústicos (directo, temprano y tardío). *Tecniacústica 2018: 49º Congreso Español de Acústica; XI Congreso Iberoamericano de Acústica; X Congreso Ibérico de Acústica*.

- **José M. Requena-Plens**, Jenaro Vera Guarinos (2018). Campo directo (útil)/reverberado (perjudicial) resultados experimentales frente a simulación en EASE. *Tecniacústica 2018: 49º Congreso Español de Acústica; XI Congreso Iberoamericano de Acústica; X Congreso Ibérico de Acústica*.
- Francisco J. Rodrigo, Pedro Poveda, Jesús Carbajo, **José M. Requena-Plens**, Jaime Ramis (2018). Comportamiento vibroacústico de contenedores cilíndricos en aire. *Tecniacústica 2018: 49º Congreso Español de Acústica; XI Congreso Iberoamericano de Acústica; X Congreso Ibérico de Acústica*.
- **José M. Requena-Plens**, Jenaro Vera Guarinos, María Soledad Yebra Calleja (2017). Campo directo (útil)/reverberado (perjudicial) resultados experimentales frente a simulación en catt-acoustic. *Tecniacústica 2017: 48º Congreso Español de Acústica; Encuentro Ibérico de Acústica; European Symposium on Underwater Acoustics Applications; European Symposium on Sustainable Building Acoustics*.

Theses & projects

- **José M. Requena-Plens** (2019). Difusores acústicos basados en resonadores de membrana y placa. *Universitat Politècnica de València. Departamento de Física Aplicada. Escuela Politécnica Superior de Gandia*.
- **José M. Requena-Plens** (2018). Estudio de la relación campo directo/reverberado; útil/perjudicial. *Universidad de Alicante. Departamento de Física, Ingeniería de Sistemas y Teoría de la Señal*.